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Isometric Torque Values About Robotic Knee Using Braided Pneumatic Actuators

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ABSTRACT

Artificial muscles such as braided pneumatic actuators (BPAs) offer many advantages for robotic systems, including high durability and strength-to-weight ratios. However, their use in robotic systems is still extremely limited, in part due to their previously poor characterization of the force, length, and pressure relationship. In this work, test setups are created to compare forces and torques produced by Festo fluidic BPAs with leading models. Predicted torque values based on developed models are compared with recorded torque data to determine the major sources of error in current modeling tools. Our analysis of the data has resulted in 1) an improved model of effective torque that includes a constant length offset and series spring elements, and 2) a correction factor for when the BPAs wrap around joints to account for the loss in usable length. Finally, we show that Festo BPAs can meet or exceed human isometric knee torque values over a majority of the knee range of motion. This work demonstrates a greatly improved model that predicts joint torques produced by Festo BPAs in biomimetic designs. This will lead to faster design processes and the development of biomimetic robots that are able to more accurately reproduce the range of motion and isometric torque profiles that exist in the animals they are mimicking.

Keywords: BPA, Biomimetic, Function Fit, PAM, Artificial Muscle, Bioinspired, Bipedal Robot, Isometric Knee Torque

1 INTRODUCTION

Biomimetic robots offer many potential applications, ranging from assisting in factories, environmental exploration, and even understanding our own bodies. By developing high-fidelity biomimetic robots, we can perform tests that help improve our understanding of both biomechanics and the underlying neuromechanical systems that control them (Shin et al., 2018; Asano et al., 2019; Ijspeert, 2020; Goldsmith et al., 2021; Hunt et al., 2017b; Schilling et al., 2022). With these robotic systems, experiments can be performed that would not be practical or ethical if done on human test subjects, such as turning off part of the neural control system or removing specific muscles. Such tests allow for direct testing of how different components of the neuromechanical system interact. Such robotic platforms offer new opportunities for

testing neural control theories that are not currently possible, and are potentially faster and cheaper than human observation studies.

Biomimetic robots built with artificial muscles provide opportunities to investigate how muscles are controlled to produce desired motions. A few such biomimetic robots exist, such as the impressive Kengoro and Kenshiro robots designed by Asano et al. (Asano et al., 2017, 2019). These robots have proportional bodies to humans, anatomically correct musculotendon arrangements, rolling-sliding joints, and joint ranges and output power that are comparable with human-measured values. The artificial muscles in these works use brushless DC motors, wires (as tendons), gears, and integrated tension measurement devices. A dog-like robot robot developed by Aschenbeck (Aschenbeck et al., 2006) and actuated with Festo BPAs, was used to show how central pattern generators provide robust locomotion patterns that are adaptive to perturbations in ground speed (Hunt et al., 2017b). Unfortunately, artificial muscles do not exactly replicate muscle properties, and these robots were not capable of capturing all the degrees of freedom and joint torques of animal limbs they mimic.

Previous research has worked to bridge this gap through the development of higher-fidelity artificial muscle models. We have developed an actuator force model force for Festo braided pneumatic actuators (BPAs) by taking data at a significant number of lengths and pressures for both $10\,\mathrm{mm}$ and $20\,\mathrm{mm}$ uninflated muscles (Bolen et al., 2026). While this model is more accurate than others for force prediction, additional correction factors need to be considered to calculate joint torque in a robotic system.

It is unknown how different attachment locations and interactions between the artificial muscles and other components affect torques produced on a robot. Previous work modeled how the location of muscle attachments affects the torque capabilities of each muscle (Bolen and Hunt, 2019). When compared with human data, it was determined that if artificial muscles are placed in the same locations as human muscles, the developed robot would not be able to achieve the same range of motion (RoM) or torque that humans achieve. To improve the range of motion and torque generated by the BPAs, optimization methods were developed that find attachment locations that can produce torque curves and ranges of motion that match human capabilities (Morrow et al., 2020). However, bending and kinking of the artificial muscle as it wraps around a joint reduce the force produced by the muscle, and this was not accounted for. Joint friction can also cause a decrease in torque, as can non-rigid elements of the artificial leg. Therefore, the work presented here also seeks to understand how much of an effect these interactions have on joint torque and develop correction factors that can improve modeling methods.

An isometric knee test stand was created to allow measurements of the torque about the knee joint in various configurations. Data collected from the test stand are compared with theoretical torque values, and is used to develop an improved model that incorporates losses due to compliance. This work also demonstrates that the improved attachment locations produced by the optimization method in previous work (Morrow et al., 2020) are capable of producing torques that match or exceed human capabilities across a wide range of joint angles.

2 METHODS AND MATERIALS

2.1 Overview

This study begins with building a jig to measure knee torque produced by BPAs at various knee angles (θ_k). With the expectation that measured torque will be less than that predicted by the model, the robot muscle arrangement and knee design are simplified to elucidate the variables that most affect the results.

Two distinct joints are used and tested: a simplified pinned knee joint and a biomimetic knee (Steele, 2018). Each joint is driven by a single flexor and a single extensor muscle. Variations were made to the muscle origin/insertion locations, resting lengths, diameters, and whether they used an artificial tendon. The results are compared with two models and the new data was used to create an updated robot model.

2.2 Actuator Force Model

The actuator force model used is fully described in Bolen et al. (2026) and is briefly reproduced here. First, the amount of contraction ε in a BPA is calculated as

$$\varepsilon = \frac{(l_0 - l_{\mathrm{m}})}{l_0} \tag{1}$$

where l_0 is uninflated resting length of the BPA measured from hose clamp to hose clamp. Current muscle length is l_{m}.

Similarly, ε_{620} is the amount of contraction in a BPA without external load when inflated at $620\,\mathrm{kPa}$, calculated as

$$\varepsilon_{620} = \frac{(l_0 - l_{620})}{l_0} \tag{2}$$

where l_{620} is the muscle length measured at $620\,\mathrm{kPa}$ and free from external loads.

The amount of relative contraction of a BPA ε^* can then be calculated by dividing equation (1) by (2), such that

$$\varepsilon^* = \frac{\varepsilon}{\varepsilon_{620}} \tag{3}$$

describes the BPA's current contraction as a normalized percentage of it's total.

The amount of normalized isometric force F^* in a Festo BPA is then calculated as

$$F^*(\varepsilon^*, P^*) = \begin{cases} c_0 \cdot \big(\exp(-c_1 \cdot \varepsilon^*) - 1\big) + P^* \cdot \exp\big(-c_2(\varepsilon^*)^2\big) & \text{for } F^* > 0 \\ 0 & \text{for } F^* \leq 0 \end{cases} \tag{4}$$

where P^* is pressure P normalized by $620\,\mathrm{kPa}$. BPA diameter dependent coefficients are listed in (Bolen et al., 2026)(p. 10).

The maximum force F_{max} in a BPA is calculated as

$$F_{max}(l_0, P_{\max}) = a_1 \cdot P \cdot \arctan\left(a_2 \cdot P \cdot (l_0 - 0.0075)\right) \tag{5}$$

with coefficients listed in (Bolen et al., 2026)(p. 7). When maximum system pressure is set to $P = 620\,\mathrm{kPa}$, $F_{\max}(l_0, 620) = F_{620}(l_0)$. Multiplying (4) by (5) yields

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0) \tag{6}$$

as the equation for isometric force F in a Festo BPA.

2.3 Isometric Knee Torque

2.3.1 Robot Model

The primary components of the robot leg are the knee joint, femur, tibia, BPA assemblies, and artificial tendons (Fig. 1**A–C**). The artificial bone components are 3D printed using a combination of Onyx, carbon fiber, and high-strength high-temperature fiberglass on Markforged Onyx One and Mark Two printers. Onyx is a proprietary Markforged material that consists of chopped carbon fiber in nylon. Artificial tendons are made with Shimano bicycle brake cable.

The leg assembly geometry and dimensions are based on bones scans of a $6\,\text{ft}$ ($1.83\,\text{m}$) tall person (Bolen and Hunt, 2019; Seth et al., 2018; Delp et al., 1990, 2007). Two knee joints were tested, one pure rotational joint, and one biomimetic joint with rotation and translation. Each joint is driven by two antagonistic Festo actuators. Both robot joints have the same center of rotation in the femur frame at $\theta_\text{k} = 0\,\text{deg}$.

Details on the biomimetic robot knee used in this study can be found in Steele (2018); Steele et al. (2017, 2018). In brief, this 1-DoF joint (Fig. 2) uses a four-bar linkage to change the effective moment arm during rotation. The intersection point of the internal crossed links define the instantaneous center of rotation (ICR). This causes the joint ICR location in the femur frame to translate backwards in the X and Y directions (Wu and Cavanagh, 1995; Wu et al., 2002) during knee rotation, similar to how the human knee behaves (Bolen and Hunt, 2019; Morrow et al., 2020; Seth et al., 2018; Delp et al., 2007, 1990). Correct location of the ICR is important for both theoretical and experimental torque calculations.

One uniarticular knee extensor and one uniarticular knee flexor are used to actuate the joint. Each knee joint type was tested using $\phi 10\,\text{mm}$ Festo BPAs. Additionally, the biomimetic knee joint was tested using a $\phi 20\,\text{mm}$ Festo BPA. Each artificial muscle was pinned at the muscle origin location on the femur while the other end was either pinned or attached to the muscle insertion location on the tibia via an artificial tendon.

Moment arms are calculated using vector components between muscle locations and the joint center of rotation. Given a vector $\vec{d}$ from a joint to the line of action of the force vector $\vec{F}$, the moment arm $\vec{r}$ is the shortest distance to this line of action. In this study, it is calculated using the method specified by Young and colleagues (Young et al., 2019). The scalar moment arm, r_k, about the z_k axis (i.e. the Z axis in the knee frame) is calculated as

$$r_k = \vec{p_{proj,xy}} \cdot \frac{\vec{p_f} \times \hat{z_k}}{\|\vec{p_f} \times \hat{z_k}\|} \tag{7}$$

where $\vec{p_{proj,xy}}$ is the free muscle segment projected onto the XY–plane, which is normal to the joint axis. $\vec{p_f}$ is the projected muscle segment vector. Moment arm is calculated in the tibial body frame.

Full details of the existing model and the method to calculate torque in the modeled robot are given din (Bolen and Hunt, 2019; Morrow et al., 2020).

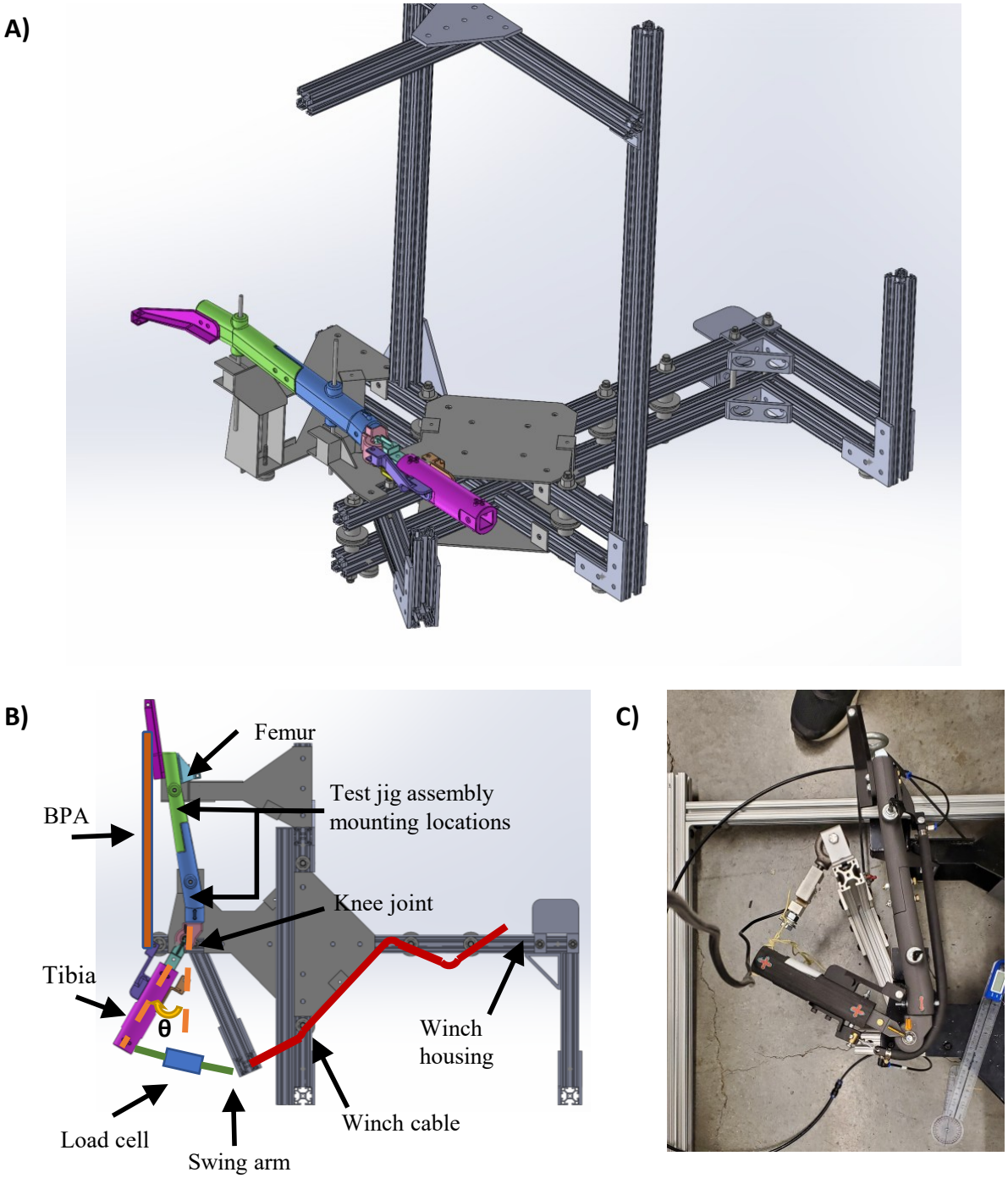


Figure 1. **(A)** CAD solid model assembly of the revolute joint knee in the test stand. **(B)** 1-DoF pinned-joint robot knee in the test apparatus with important components labeled. Leg is shown with 30 degrees flexion, i.e. $\theta_k = -30°$. **(C)** Picture from isometric torque testing. Setup shows the revolute knee configured for a test of the extensor BPA. The knee is flexed with $\theta_k = -120°$. Also note the compressed shape of the inflated BPA on the knee during this high degree of flexion.

2.3.2 Knee Torque Data Collection and Equipment

A test jig was built to collect knee torque measurements at different knee angles over its range of motion. The test stand frame is made predominantly out of 80/20™ 1515 series extruded aluminum components.

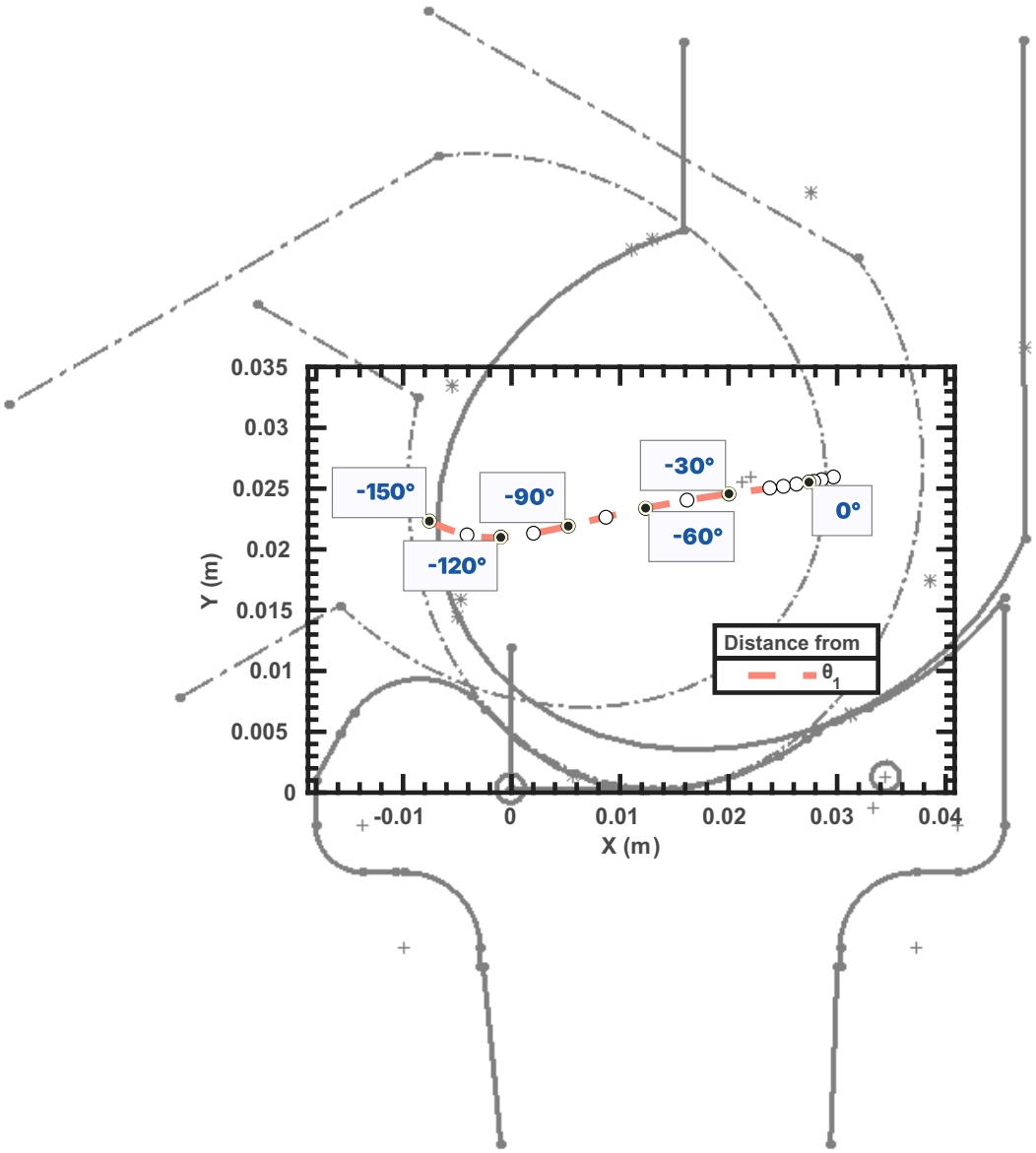


Figure 2. ICR for biomimetic robot knee as a function of knee angle. Measured from θ_1 on the tibia. Plot is overlayed on model of the knee at $\theta_k = 0°$ (solid line), and $-60°$ and $-120°$ (dotted dash lines). White circles represent measured points in $-15°$ increments from $-150°$ to $-15°$, $-5°$ increments from $-15°$ to $0°$, and then at $2°$, $5°$, and $10°$.

The knee joint is free to rotate the tibia while the femur is fixed to the frame. A force sensor has one end connected to the tibia and the other is connected to a swing arm. The swing arm is tied to a winch with $3/16$ inch Kevlar rope from Quality™ Nylon Rope routed over intermediate idler pulleys (see Fig. 1**B** and 1**C**). At each measured position of θ_k the winch would be locked and the BPA would be inflated and then deflated. This test procedure kept the measured results as isometric as possible. The knee joint is co-axial with gravity to eliminate gravitational torque (see Fig. 1**A** and Fig. 1**C**).

Force data was collected using one of two different sensors. The first is a MODERN STEP $300\,\mathrm{kg}$ digital crane scale. The second force sensor is a CALT DYLY-103 $100\,\mathrm{kg}$ S shaped load cell. The load cell is used in conjunction with a HX711 Load Cell Amplifier. Pressure is measured with a Freescale MPX5700 GP

$5\,\mathrm{V}$ pressure sensor. Building air supply pressure is controlled with two pressure regulators in series. The first is a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second is a Husky $3/8$ inch High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-T532S-8K valve manifold was used to supply air to the actuator. Pressure and load cell amplifier data are sent to Matlab via an Arduino Uno style Sparkfun BlackBoard C microcontroller.

The length measurements were made using a tape measure. When there was sufficient curvature in the BPA during knee extensor torque tests, a flexible piece of string was used to mark the axial length and then transferred to the tape measure. Angle measurements were taken with a Medigauge digital electronic goniometer or with MALENOO analog goniometers of 6, 8, or 12 inch lengths.

2.3.3 Hybrid Torque Calculation Method

A Hybrid Torque calculation method was developed to help uncover inconsistencies between the model and recorded torque values. This method measures the actual muscle length and angles of the robot to calculate force and joint torque. With this method, it is easier to determine which parts of the torque equation did not achieve their predicted value and provide a correction to that part of the equation. The classical mechanics way to calculate the torque $\vec{M}$, about a joint is to take the cross product of distance, $\vec{d}$, and force, $\vec{F}$

$$\vec{M} = \vec{d} \times \vec{F} \tag{8}$$

where distance $\vec{d}$ and force $\vec{F}$ are described in the previous section. Knee torque is the scalar torque M about the Z axis, which can simplify Eq. 8 to

$$M = r \times F \tag{9}$$

where r is the moment arm measured by a tape measure, and F is the scalar force, calculated by measuring the BPA's P, l_0, l_{620} and l_{m} values and using the model fit equations determined in the following section.

3 RESULTS

3.1 Flexor on Pin Joint

Experimentally measured torque is compared with various prediction methods while using flexor BPAs on the pinned knee joint (Fig. 3). These results show that the original model and the improved BPA force model overpredict the measured torque at the knee for all joint angles, although the improved model performs slightly better (Fig. 3**A** and3**B**). However, when the measured parameters (l_{m} and r_k) are used in the hybrid model, then predictions closely match the experimental data. This indicates that the kinematics of the torque model are incorrectly predicting l_{m} and/or r_k.

Analysis of measured muscle length data shows that it was consistently shorter than the predicted muscle length at all joint angles. This shorter length results in less muscle force, a shifted RoM, and less torque in the experimental setup. That this shorter length occurs when $\varepsilon = 1$ and $F = 0$ indicates that a constant length offset should be used. Therefore, to produce an improved torque prediction model, l_{m} was recalculated as

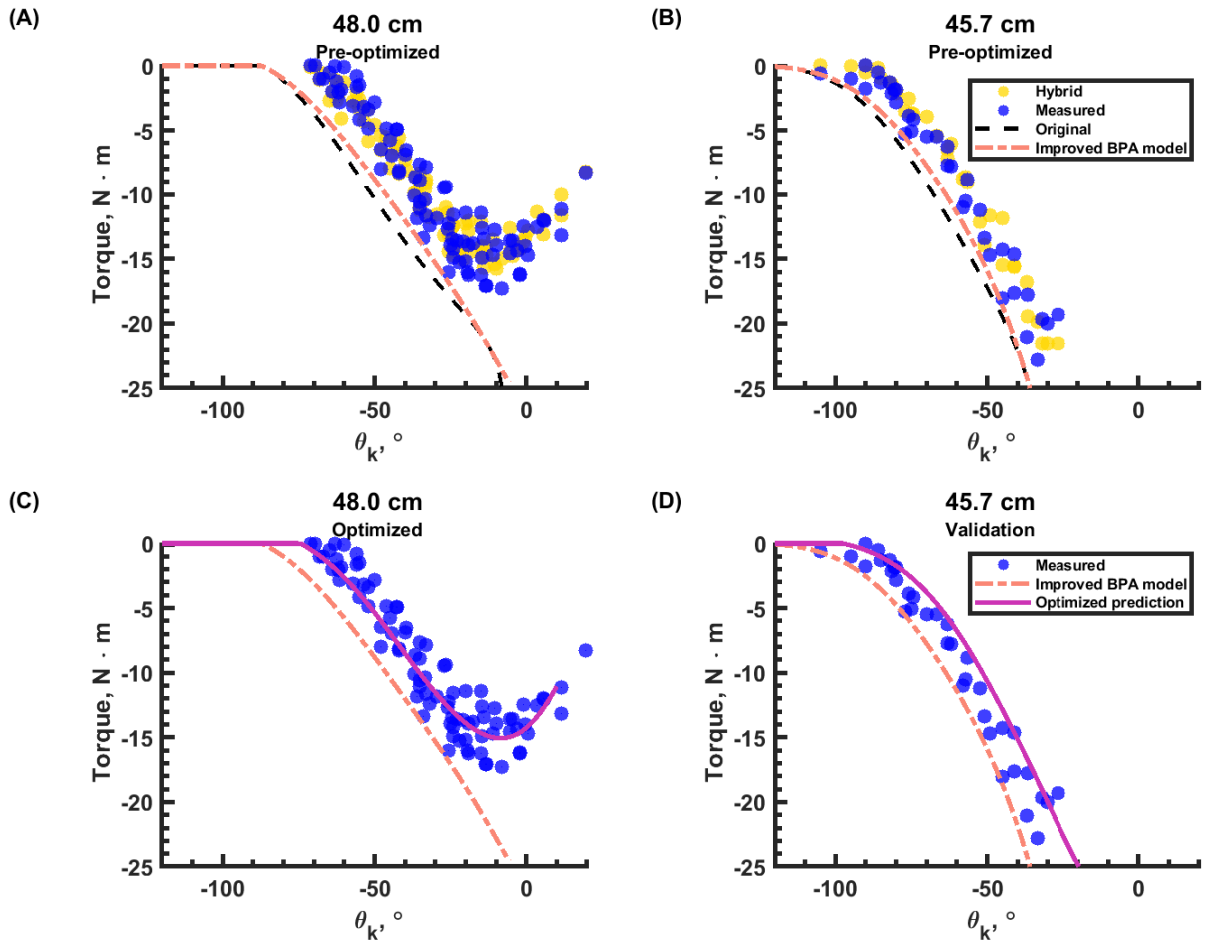


Figure 3. Results for the pinned knee joint using flexor BPAs. Dashed black line shows the torque predicted with our original model. Blue dots are the experimentally measured torque. Yellow dots represent the hybrid method of calculating torque. The magenta line represents the predicted torque. **(A)** Measured and pre-optimized torque prediction for the $48.5\,\mathrm{cm}$ BPA. **(B)** Measured and pre-optimized torque prediction for the $l_0 = 45.7\,\mathrm{cm}$ BPA. **(C)** Theoretical and measured torque for the $l_0 = 48.5\,\mathrm{cm}$ BPA using optimization results. **(D)** Predicted measured torque for the $l_0 = 45.7\,\mathrm{cm}$ BPA using the optimization results

$$l_{\mathrm{m}} = l_{LMT} - 2\,l_{fitting} - \chi_0 \tag{10}$$

where l_{LMT} is the path length of the musculotendon from origin to insertion point, $l_{fitting}$ is the length of the end caps, and χ_0 is a constant length offset.

Additionally, as the muscles produce more force, compliant components in the limb segments and attachment hardware deform. This deformation changes the effective braided pneumatic actuator (BPA) length, which alters contraction and therefore changes the force predicted by the BPA model. This effect is most apparent at high forces and low contractions. To correct this error, we approximated the combined hardware compliance using Hooke's law in the bracket's principal directions.

A local bracket frame was established at the attachment bracket. The $\hat{x}_{br}$ direction was aligned with the vector from the bracket reference point to the BPA attachment point and represents the stiffest direction of

the bracket. The $\hat{y}_{br}$ direction was chosen to point in the posterior direction with respect to the tibia frame, corresponding to the primary cantilever bending direction (see Fig. 4), while the $\hat{z}_{br}$ direction completes the right-handed frame. The bracket stiffness matrix is modeled as a diagonal matrix to reflect independent stiffness components along its principal axes:

$$\mathbf{K}_{br} = \begin{bmatrix} \chi_1 & 0 & 0 \\ 0 & \chi_2 & 0 \\ 0 & 0 & \chi_1 \end{bmatrix} \tag{11}$$

The corresponding local compliance matrix is

$$\mathbf{C}_{br} = \mathbf{K}_{br}^{-1} \tag{12}$$

Force that doesn't account for deformation is $\mathbf{F}$. The force applied to the bracket was expressed in the bracket frame as $\mathbf{F}'$. This force vector can be written in terms of its magnitude and unit direction as

$$\mathbf{F}' = \left\|\mathbf{F}'\right\| \hat{\mathbf{u}} \tag{13}$$

where $\hat{\mathbf{u}} = [u_x, u_y, u_z]^T$ is the BPA force direction expressed in the bracket frame. The bracket deflection vector, $\boldsymbol{\epsilon}_{br}$, was calculated from Hooke's law as

$$\mathbf{F}' = \mathbf{K}_{br}\, \boldsymbol{\epsilon}_{br} \tag{14}$$

Solving Eq. 14 for the bracket deflection gives

$$\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\, \mathbf{F}' = \mathbf{C}_{br}\, \mathbf{F}' \tag{15}$$

Substituting Eq. 13 into Eq. 15 gives

$$\boldsymbol{\epsilon}_{br} = \left\|\mathbf{F}'\right\| \mathbf{K}_{br}^{-1}\, \hat{\mathbf{u}} \tag{16}$$

For the diagonal stiffness matrix in Eq. 11, Eq. 15 is equivalent to the component-wise expression

$$\boldsymbol{\epsilon}_{br} = \begin{bmatrix} F'_{x,br}/\chi_1 \\ F'_{y,br}/\chi_2 \\ F'_{z,br}/\chi_1 \end{bmatrix} \tag{17}$$

Although $\boldsymbol{\epsilon}_{br}$ is a three-dimensional deflection vector, the BPA force model depends on the scalar change in effective BPA length (δ). Therefore, only the component of $\boldsymbol{\epsilon}_{br}$ along the BPA line of action directly changes the effective BPA length. This scalar bracket deflection is the projection of $\boldsymbol{\epsilon}_{br}$ onto $\hat{\mathbf{u}}$:

$$\delta_{br} = \hat{\mathbf{u}}^T \boldsymbol{\epsilon}_{br} \tag{18}$$

Substituting Eq. 15 into Eq. 18 gives

$$\delta_{br} = \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \mathbf{F}' \tag{19}$$

Using Eq. 13, Eq. 19 becomes

$$\delta_{br} = \left\|\mathbf{F}'\right\| \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \tag{20}$$

The scalar stiffness along the BPA line of action due to the bracket compliance is therefore

$$k_{\hat{u},br} = \frac{\|\mathbf{F}'\|}{\delta_{br}} = \left(\hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}}\right)^{-1} \tag{21}$$

This projected-compliance form is used because the force is applied along $\hat{\mathbf{u}}$, while the bracket is allowed to deflect independently along its local $\hat{x}$, $\hat{y}$, and $\hat{z}$ directions. The reduced-order Hooke's law expression for the bracket deflection along the BPA line of action is therefore

$$\left\|\mathbf{F}'\right\| = k_{eq}\, \delta \tag{22}$$

where k_{eq} is calculated with (21) when there is no artificial tendon. When an artificial tendon was used, it was modeled as an axial spring with stiffness k_{ten}. Since both bracket deflection and tendon stretch reduce the effective BPA length along the actuator line of action, their scalar compliances were added in series:

$$k_{eq} = \left(\frac{1}{k_{br}} + \frac{1}{k_{ten}}\right)^{-1} \tag{23}$$

Since δ is the scalar compliance displacement along the BPA line of action, then the compliance-corrected active BPA length used in the force balance is

$$l_m(\delta) = l_m^{(0)} - \delta \tag{24}$$

where $l_m^{(0)}$ is calculated from (10).

The corresponding relative contraction is $\varepsilon^*(\delta)$. Since the BPA force depends on contraction, and contraction depends on the compliance displacement, δ was found using the Newton-Raphson method to solve the nonlinear force-balance equation

$$F_{\max} \cdot F_{\mathrm{BPA}}(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0 \tag{25}$$

After solving Eq. 25, the corresponding force magnitude was used in Eq. 15 to compute the vector bracket deflection. This deflection was transformed back into the limb reference frame and used to update the BPA attachment location. The routed BPA length, force direction, moment arm, and torque were then recalculated using the updated attachment location. In the present implementation, the optimized stiffness terms χ_1 and χ_2 were treated as effective compliance parameters for the attachment system rather than direct measurements of an isolated bracket. This allowed unmodeled compliance in the surrounding hardware to be represented by the reduced-order stiffness model.

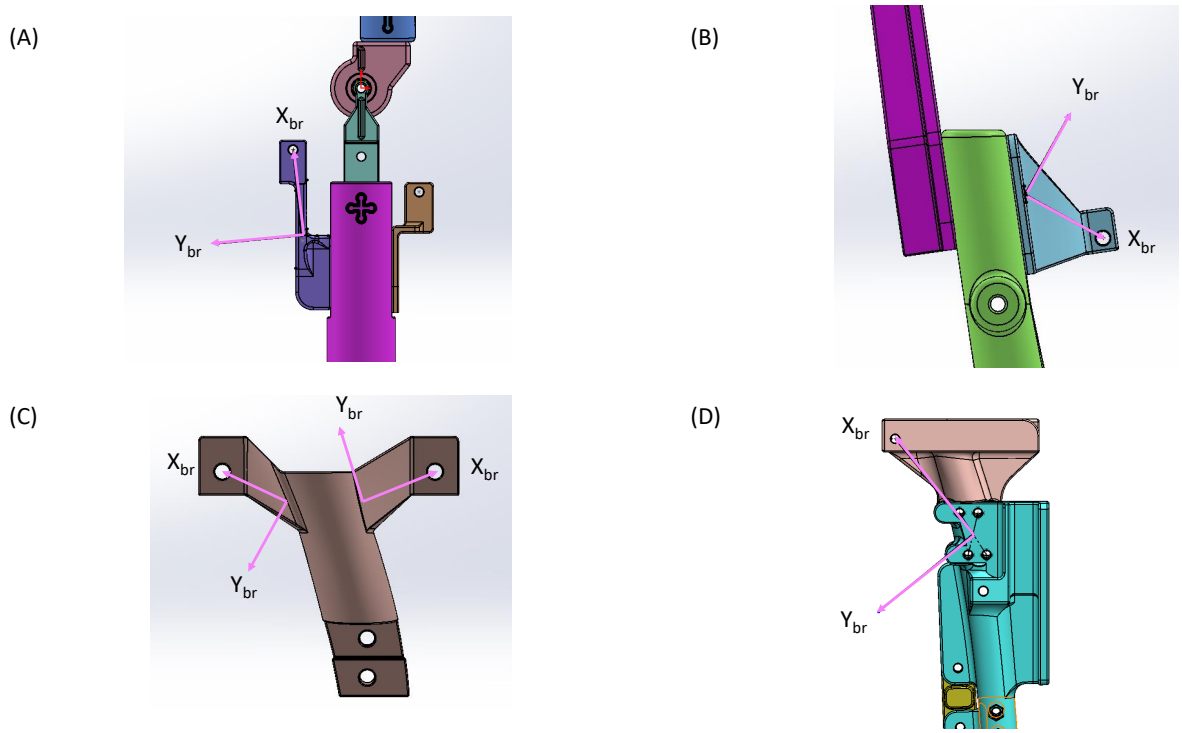


Figure 4. Bracket frame orientations for: **(A)** Flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with $\phi 10\,\mathrm{mm}$ BPAs, and **(A)** flexor configuration on the biomimetic knee with the $\phi 20\,\mathrm{mm}$ BPA.

Multiobjective optimization using gamultiobj in Matlab was used to determine the length offset (χ_0) and stiffness terms (χ_1 and χ_2) until the error between the the model and experimental data was minimized for the $l_0 = 48.5\,\mathrm{cm}$ BPA (Fig. 3C). The solution yielded $\chi_0 = 11.9\,\mathrm{mm}$, $\chi_1 = 102\,\mathrm{kN\,m^{-1}}$, and $\chi_2 = 10.6\,\mathrm{kN\,m^{-1}}$. It can be seen that the model with these correction factors greatly improves the ability to predict the joint torque, even capturing the reduction in force at small joint angles that was completely missing from the previous model. The $l_0 = 45.7\,\mathrm{cm}$ BPA was used for validation (Fig. 3D). It can be seen that the improved model does a good job of predicting the torque with a different muscle length configured to operate over a different joint range of motion.

3.2 Extensor on Pin Joint

When the new stiffness and length offet model was used to predict torque for the knee extensor, measured torque was slightly higher than the original model calculates for the $48\,\mathrm{cm}$ resting length BPA, while it was slightly lower than expected in the $46\,\mathrm{cm}$ BPA, and it was much lower in the $42\,\mathrm{cm}$. Only the K_1 and K_2 terms from the previous optimization results were used with the extensor configuration. Firstly, by inspection, the zero point for the measured torque was shifted to the right of the original model, indicating an offset that was at least opposite in sign from the χ_0 term in the previous section. Secondly, if we look at the BPA $l_0 = 46\,\mathrm{cm}$ as an example, we see good correlation between the hybrid-method calculated torque and the originally predicted torque, but a divergence in the actual measured torque.

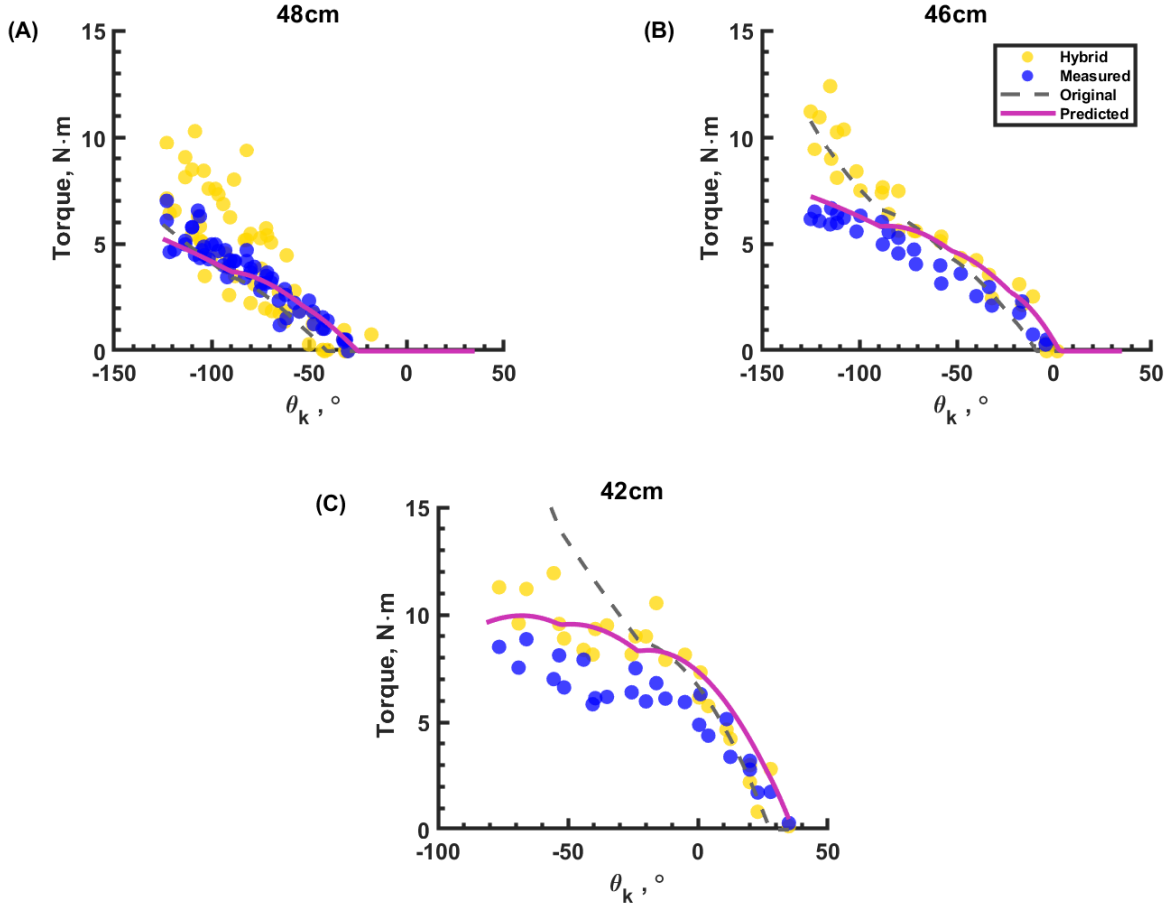


Figure 5. Pinned knee torque versus knee angle with the extensor BPAs. Blue dots are experimentally measured data, yellow dots are hybrid calculated torque, black dashed line is the original predicted values, and magenta line is the optimized predicted torque. For resting lengths, **(A)** $48.0\,\mathrm{cm}$, **(B)** $45.7\,\mathrm{cm}$, **(C)** and $41.5\,\mathrm{cm}$.

This indicates that the source of error was not as much bracket deformation, but a phenomenon that occurs as the BPA undergoes bending over the joint. The inner side of the BPA experiences less tensile stress than the outside, creating a force imbalance and a loss of part of the arc length of the BPA as usable length. We also noticed that the more contracted a BPA was over the RoM, the less the torque results were

affected by wrapping. We hypothesized that the an additional optimization term χ_3 should be introduced, such that

$$\Delta l = \begin{cases} \chi_3 \; R \; |\theta_{\text{wrap}} - \theta_{\text{k}}| \; \xi^2 & \theta_{\text{k}} \leq \theta_{\text{wrap}} \\ 0 & \theta_{\text{k}} \geq \theta_{\text{wrap}} \end{cases} \tag{26}$$

where Δl is the change in usable length of the BPA, R is the wrapping radius. θ_{wrap} is the knee flexion angle when the BPA starts to bend over the knee joint, determined from solid modeling. ξ is the additive complement to relative strain, such that

$$\xi = 1 - \varepsilon^* \tag{27}$$

Combining (27) into (26), and subtracting Δl from (10) yields

$$l_{\text{m}} = l_{LMT} - 2 \, l_{fitting} - \chi_0 - \Delta l \tag{28}$$

The optimization algorithm was again gamultobj in Matlab. This time, the muscle origins bracket was chosen. The bracket frame is shown in Fig. 4**B**. Since the bracket forms an arch shape and extends for $75\,\text{mm}$ in the Z direction, for this configuration only we set K_2 in the X_{br} direction, and K_1 in the Y_{br} and Z_{br} directions. The optimization algorithm for this set solves for the best average goodness-of-fit (GoF) measures RMSE, FVU, and maximum absolute residual for two BPAs, while the third one was held out. Three independent rounds of optimization were performed, so that one of each of the BPAs could be held out as validation. Then we picked the solution with the shortest distance between the average normalized GoF and the validation data, while also requiring improved GoF values for all the individual BPAs, with the exception of maximum absolute residual on the $l_0 = 48.0\,\text{cm}$ BPA.

Results are shown in Fig. 5. The solution was found while holding the $l_0 = 41.5\,\text{cm}$ BPA for validation. The optimization resulted in $\chi_0 = -10.9\,\text{mm}$ and $\chi_3 = 0.123$. It should be noted here that the length offset χ_0 is almost equal in magnitude, but opposite sign, as the flexor configuration. The goodness of fit values for the torque prediction is found in table **??**.

3.3 Extensor on Biomimetic Knee

We obtained results using the biomimetic knee with a $\phi 10\,\text{mm}$ BPA in the extensor configuration. The bracket frame origin is the centroid of the cantilevered segment that holds the muscle origin bracket to the proximal femur (see Fig. 4**C**, right side).

The torque results for a $l_0 = 52.0\,\text{cm}$ BPA are shown in Fig. 6. The torque values measured are significantly smaller than both the original prediction and back-calculated torque using the hybrid method. The results from the previous extensor configuration lead to a better torque prediction. GoF values can be found in table **??**. Fig.7 shows the difficulty of modeling a BPA path that passes through the femoral condyles. Using the optimization results for extensor BPAs on the pinned knee joint, the $l_0 = 52.0\,\text{cm}$ BPA has an RMSE $= 0.8920$, FVU $= 0.3396$, and a maximum absolute residual of 1.4188.

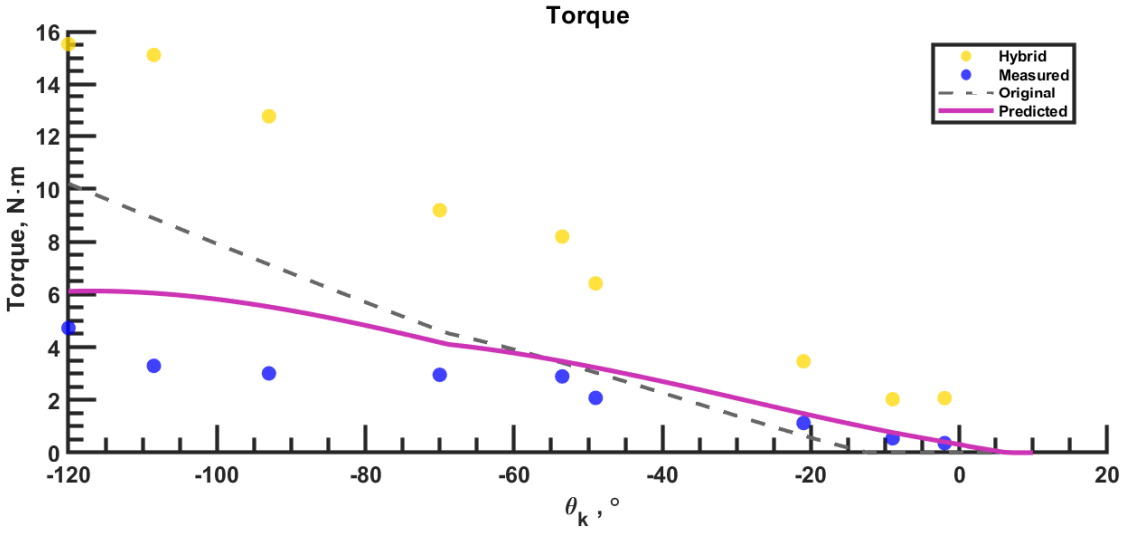


Figure 6. Measured versus expected isometric torque results using a $\phi 10\,\mathrm{mm}$, $l_0 = 51.8\,\mathrm{cm}$ extensor BPA on the biomimetic knee. Measured results are blue dots. Original prediction is a dashed black line. Optimized prediction is a solid magenta line. Hybrid calculation shown in yellow dots.

3.4 Flexor on Biomimetic Knee

Fig. 8 shows the isometric torque values of the biomimetic knee from measurements, the Hunt model (Hunt et al., 2017a), and the predictions of the updated model. The values for χ_0, χ_1, and χ_2 derived from the flexors on the pinned knee were used. The bracket frame for the $\phi 10\,\mathrm{mm}$ BPA is shown on the left hand side of Fig. 4**C**, and for the $\phi 20\,\mathrm{mm}$ it is 4**D**. Fig. 8**A** represents a $\phi 10\,\mathrm{mm}$ $l_0 = 41.5cm$ BPA with a $21\,\mathrm{mm}$ artificial tendon made out of $\phi 1.5\,\mathrm{mm}$ Shimano bicycle brake cable. The brake cable was wrapped four times. The BPA was pressurized to $604\,\mathrm{kPa}$. The predicted values have an RMSE $= 0.3007$, FVU $= 0.0102$, and a maximum absolute residual of $0.4647\ N \cdot m$. Fig. 8**B** represents a $\phi 20\,\mathrm{mm}$ $l_0 = 41.5cm$ BPA with a $15\,\mathrm{mm}$ artificial tendon made out of Shimano bicycle brake cable (wrapped six times). Maximum human isometric torque values are shown as the black dotted line. The experiment was performed at $620\,\mathrm{kPa}$ and $325\,\mathrm{kPa}$. At $620\,\mathrm{kPa}$, the predicted values have an RMSE $= 1.8024$, FVU $= 0.0499$, and a maximum absolute residual of $3.5678\ N \cdot m$. The BPA torque does not exceed the magnitude of the human torque from ca. 60 – 100 degrees knee flexion. Allowing for a $\pm 20\%$ deviation from the human mean, the BPA does not produce human levels of torque from 78 – 103 degrees knee flexion.

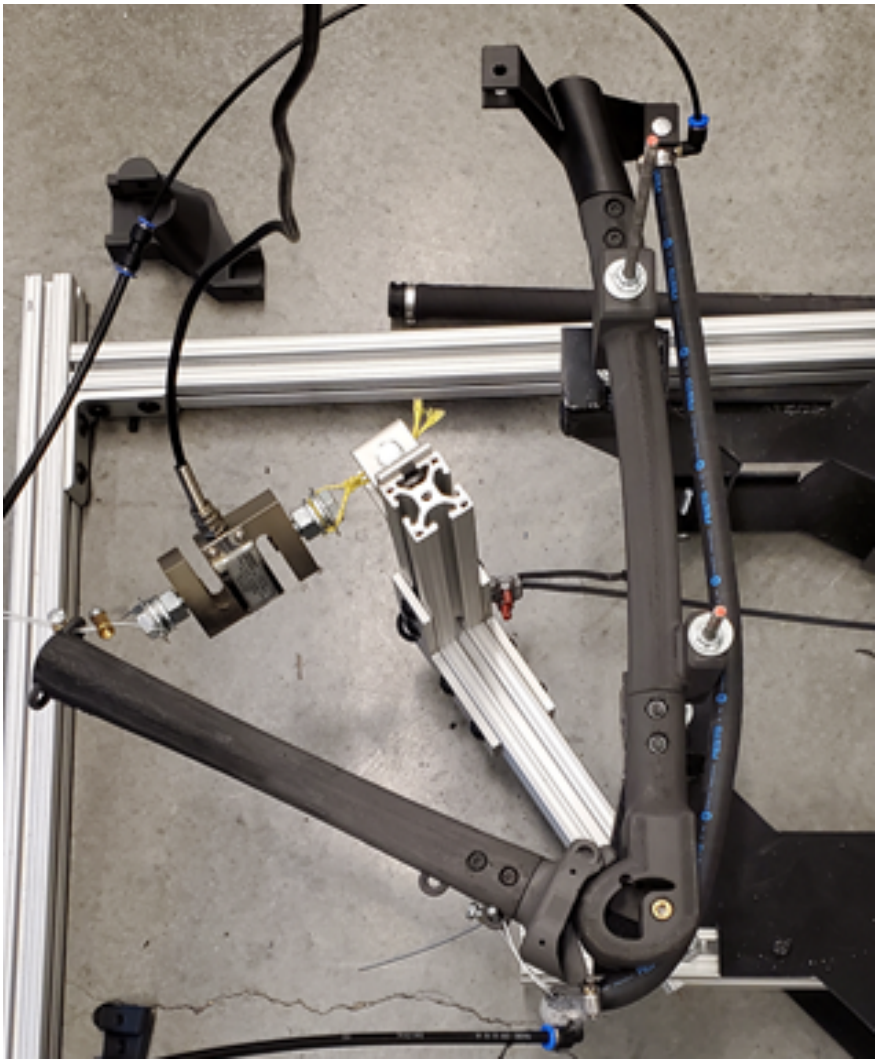


Figure 7. A configuration that is particularly hard to model. The BPA passes between the femoral condyles close to ICR.

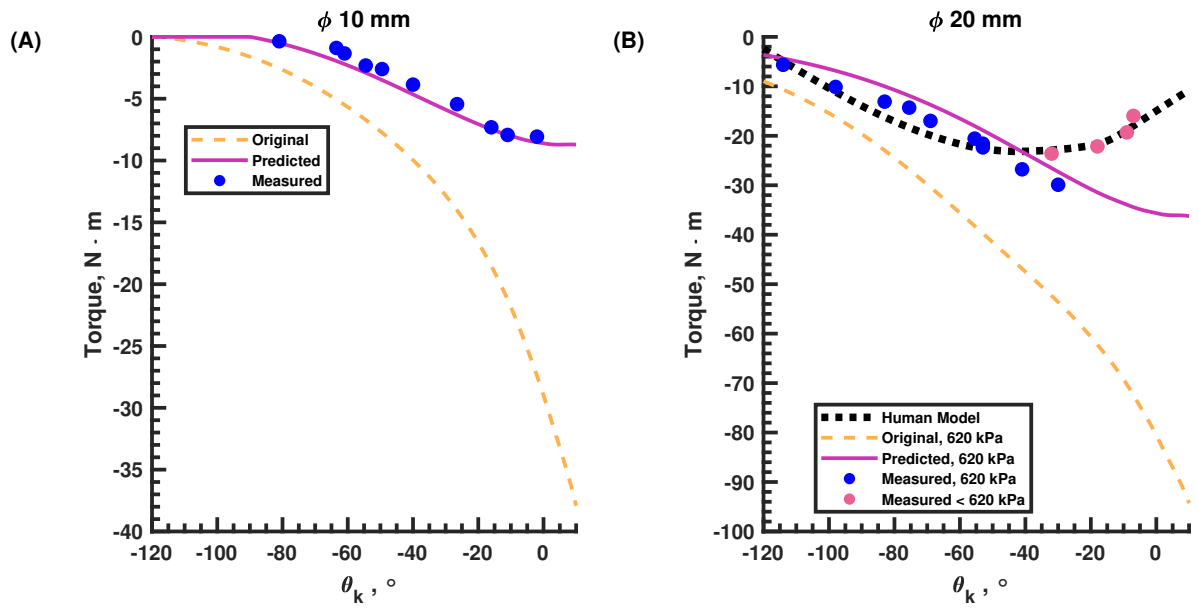


Figure 8. Biomimetic knee torque for **(A)** $\phi 10\,\mathrm{mm}$ $l_0 = 41.5\,\mathrm{cm}$ flexor BPA with a $21\,\mathrm{mm}$ artificial tendon. Blue dots show the measured torque values. Magenta line is the predicted torque using the improved BPA characterization and optimization from the previous sections. The dashed light orange line shows the torque predicted using only the improved BPA characterization. **B)** $\phi 20\,\mathrm{mm}$ $l_0 = 41.5\,\mathrm{cm}$ flexor BPA with a $15\,\mathrm{mm}$ artificial tendon uses the same colors for $620\,\mathrm{kPa}$ results. Human values are given as a black dotted line. Due to the BPA exceeding the ultimate strength of the femur, torque measurements were taken at $P < P_{620}$ to show the desired RoM is acheivable. Results for torque measurements at $560\,\mathrm{kPa}$, $421\,\mathrm{kPa}$, $325\,\mathrm{kPa}$ and $281\,\mathrm{kPa}$ (from left to right, respectively), are shown as pink dots.

4 DISCUSSION

Our improved BPA characterization predicts torque more accurately than the original BPA model (see Fig. 3**A** and 3**B**). This only accounts for part of the error. There are certainly many factors that affect the torque results. Compliance in the testing system, position location tolerance, flexibility of brackets, imperfect path modeling, and variation in individual BPAs are other factors.

Simplifying the model and testing it allowed us to see how we were deficient in our previous analysis. The isometric system is not rigid. There is compliance in the winch cable, artificial tendons, and brackets that bend and stretch. We introduced a constant length offset term χ_0, axial stiffness χ_1, and bending stiffness χ_2. The bracket frame of reference was placed where it starts to cantilever and is rotated about Z. Optimization found results for a $\phi 10\,\mathrm{mm}$ BPA with $l_0 = 48\,\mathrm{cm}$ in the simple pinned knee configuration. These results were validated with second $\phi 10\,\mathrm{mm}$ BPA with $l_0 = 45.7\,\mathrm{cm}$.

The modeling of extensor BPA paths that wrap around the knee joint is an approximation with inherent error (Fig. 7). When the moment arm is short, small changes to it can have a large effect on the resultant torque (Young et al., 2019). It was assumed that wrapping the BPA around the joint would be easier to account for than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction. During greater magnitude knee flexion angles, shorter l_0 BPAs were noted as being stretched and compressed (see Fig. 1**C**). The same force that caused the stretch and compression would also sometimes cause the BPA to slip off of the anterior bolt head of the bolt that holds the femur to the knee. This would result in a $\pm$ 20 mm displacement in the Z direction, which means a change the force vector and a change in torque. The system designer, too, must be mindful. Fig. 5 shows a deviation between measured and originally predicted results that widens as expected torque increases. This indicates system compliance. However, the hybrid calculated torque matching or exceeding predicted torque, except for higher degrees of flexion in the $l_0 = 42\,\mathrm{mm}$, indicates a loss of functional l_m. BPA This is why the χ_3 term was created in (26). Fig. 5 shows the improved fit from running the optimization that includes this term.

The biomimetic knee joint with an extensor BPA was used to validate the extensor optimization results. The predicted line more closely matches the torque values. However, the predicted torque is still higher than experimentally measured torque.

In the presented nonlinear biomimetic robotic system, rigid body mechanics simplifications, often encountered in other robotic and biomechanical systems, are removed. Instead, working from the simple to the complex elucidated mechanisms of system compliance that will lead to much more accurate first iteration designs. Technologies have been developed to predict system bending and axial compliance, a constant length offset, and a loss of usable length that occurs as artificial muscle bends around a joint.

In equation $m\ddot{\varepsilon}^* + c(\dot{\varepsilon}^*)\dot{\varepsilon}^* + k(\varepsilon^*)\varepsilon^* = F$ given $k(\varepsilon^*, P^*) = \frac{\partial F}{\partial \varepsilon^*}\big|_{P^*} = -c_0 c_1 e^{-c_1\varepsilon^*} - 2c_2\varepsilon^* P^* e^{-c_2(\varepsilon^*)^2}$ then if we set mass m and external force F to zero we get $c(\dot{\varepsilon}^*) = (k(\varepsilon^*)\varepsilon^*)/\dot{\varepsilon}^*$

RESOURCE IDENTIFICATION INITIATIVE

- MATLAB (RRID:SCR_001622)
- OpenSim (RRID:SCR_002683)
- Arduino UNO (RRID:SCR_017284)

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analysis, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

AUTHOR CONTRIBUTIONS

Conceptualization, B.B. and C.M.; Methodology, B.B., C.M., and A.H.; Software, B.B., C.M., and L.B.; Validation, B.B. and A.H.; Formal Analysis, B.B.; Investigation, B.B., C.M., and C.M.; Resources, B.B., L.B., and A.H.; Data Curation, B.B.; Writing – Original Draft Preparation, B.B.; Writing – Review & Editing, B.B., C.M., L.B., and A.H.; Visualization, B.B. and A.H.; Supervision, B.B. and A.H.; Project Administration, B.B. and A.H.; Funding Acquisition, A.H.

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SUPPLEMENTAL DATA

Supplemental Data includes figures for the test setup.

DATA AVAILABILITY STATEMENT

The data sets are available from the authors upon reasonable request.

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FIGURES

TABLES

Table 1. Optimized coefficient values for flexor and extensor configurations. X_0 is a constant length offset (in meters), X_1 and X_2 are stiffness values (in N/m) in the bracket's X and Y directions, respectively, and X_3 is the unitless wrapping factor.

	$\mathbf{X}_0$	$\mathbf{X}_1$	$\mathbf{X}_2$	$\mathbf{X}_3$
Flexor	0.0119	1.02E+05	1.06E+04	–
Extensor	-0.0109	1.02E+05	1.06E+04	0.123